

Shivaji Chaulagain

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EDUCATION

St. Xavier's College, Tribhuvan University

Bachelor of Science in Physics

Average percentage: 74.50 (CGPA: 3.79/4.00 Scholaro)

Kathmandu, Nepal

2019–2024

RESEARCH EXPERIENCE

Comparative Study of Soft and Hard Boundary Constraints in Physics-Informed Neural Networks for Quantum Mechanical Eigenvalue Problems | [DOI link](#)

Submitted

Independent Research Project, (Heliyon, Elsevier)

Jun 2025 - Sep 2025

- Initiated and coordinated a study with seven peers to investigate how boundary condition enforcement techniques affect solution accuracy and convergence in PINNs.
- Compared the performance of soft- and hard-constrained models in solving the Schrodinger equation for various potentials, benchmarking against analytical and numerical solutions.
- Demonstrated that hard-constrained models with ansätze reflecting the required asymptotic behavior and a variable parameter converge faster and achieve higher accuracy.
- Achieved over 100× improvement in accuracy using adaptive learning rates for hard-constrained models compared to constant rates.
- Co-authored a manuscript detailing methodology and results.

A Comparative Data Analysis of Gravitational Wave Signals | [Thesis link](#)
From Binary Star Coalescences of GW170817, GW190814 and GW190521

Undergraduate Thesis

Supervisor: Dr. Shreeram Nagarkoti, St. Xavier's College

Apr 2024 - Sep 2024

- Analyzed gravitational wave data from GW170817, GW190521, and GW190814.
- Created strain data plots, ASD plots, Q-transform visualizations, match filtering, waveform generation and template matching.
- Verified that the gravitational waveforms seen in occurrences such as GW190521, and GW190814 are consistent with General Relativity's waveform predictions

TECHNICAL PROJECTS

Galaxies & AGN Multi-Survey Explorer | [App link](#)

Astronomical Research and Independent Project

- A comprehensive Streamlit web application integrates data from multiple surveys to provide a unified platform for galaxy and AGN analysis.
- Key Features:**
 - Multi-Survey Integration: DESI, SDSS, MAST, Legacy Survey, Gaia
 - Spectral Analysis, Emission line fitting
 - BPT Classification: Distinguish star-forming galaxies, AGN, and composites
 - Interactive Visualizations: Explore image data with meijering Filter, sato Filter and different advanced features analysis

Cosmic Microwave Background Explorer using CAMB | [App link](#)

Astronomical Research and Independent Project

- An interactive web application for exploring Cosmic Microwave Background (CMB) power spectra using CAMB with adjustable cosmological parameters
- Key Features:**
 - Implemented support for 21 cosmological parameters including the core Λ CDM set and advanced extensions
 - Visualize CMB Power Spectra (TT , EE , BB) and the Matter Power Spectrum $P(k)$ in real-time.
 - Enable seamless CSV export of spectra and TXT parameter export for research use

NovaScope | [App link](#)

Data Visualization Project

- A Python-based interactive dashboard for visualizing scientific datasets with drag-and-drop upload and instant plot generation using pandas, Astropy, h5py
- **Key Features:**
 - Eliminate custom scripting, enabling immediate conversion of raw scientific datasets into clean, interactive plots.
 - Robust data pipeline to handle diverse formats, supporting direct upload and parsing of CSV, TXT, DAT, NPZ, FITS, and HDF5 files.
 - Line, scatter, bar, Pdf, KDE, histogram, Regression and 3D plots, updated immediately upon upload and selection.

Generalized Aubry–André–Harper (GAAH) Research Dashboard | [App link](#)

Condensed Matter Physics- Independent Research and Visualization Project

- This dashboard is part of my independent research project “Machine Learning Framework for Time-Resolved Mobility Edges in 1D Quasiperiodic Systems.” It serves as a visualization and analysis tool for simulation data generated by a custom hybrid-parallel pipeline.
- **Key Features:**
 - focuses on the GAAH model, which exhibits a unique energy-dependent mobility edge separating localized and extended states.
 - layout covering energy spectrum, mobility edges, fractal dimensions, and global phase diagrams
 - High-performance Plotly charts and export capabilities (PNG/JSON)
 - Automated phase classification based on level spacing ratios, MSD trend diagnosis, and fractal dimension estimation

Convolutional Neural Networks Visualization App | [Github](#)

Research and Visualization Project

- This project is an interactive application for understanding Convolutional Neural Networks (CNNs), which provides visualizations, comparisons, and hands-on demonstrations of CNNs implemented from scratch and using popular deep learning libraries like TensorFlow and PyTorch.

Nepal Foreign Trade Analytics Dashboard | [App link](#)

Visualization and Analysis Project

- This dashboard shows Nepal's foreign trade from fiscal year 2014/15 to first 3 month of 2025/26.
- It explains how much Nepal imported and exported each year, the difference between them (trade balance), and changes in major products.

Demonstration of Astronomical Software Tools Project | [Github](#)

Astronomical Python Software

- Provides a comprehensive list of software tools and libraries that are essential for astronomical research and data analysis

Exoplanet Data Dashboard: Built with Astroquery and Streamlit using data from the NASA Exoplanet Archive | [Github](#)

PROFESSIONAL DEVELOPMENT

2024 Aspire Leaders Program.....

The Aspire Leaders Program | [Certificate link](#)

Participant (online)

August 2024

- Emerged with increased self-confidence, sharpened critical thinking skills, refined communication skills, and a desire to create a social impact among a global community.

ONLINE COURSES.....

13-18 April 2023: Introduction to Quantum Computing, St. Xavier's Physics Council | [link](#)

Feb 28 2024: Machine Learning with Python, Coursera | [link](#)

March 2 2024: Data Analysis with Python, Coursera | [link](#)

March 15 2024: Introduction to Deep Learning and Neural Networks with Keras, Coursera | [link](#)

July 28 2024: Advanced Learning Algorithm, DeepLearning.AI (Coursera) | [link](#)

2024: Data-driven Astronomy, The University of Sydney (coursera), | [link](#)

WORKSHOPS.....

8-9 April 2023: *Workshop on Remote Sensing in Climate Sciences (RSCS2023)*, Tribhuvan University in

cooperation with UGC Nepal, St. Xavier's College, Kathmandu and Nepal Physical Society

18-19 June, 2023: Hands-on Workshop on *Latex Training*, St. Xavier's College, Kathmandu

9-11 August 2023: *A mini-workshop on particle physics and science communication with Dr. Suyog Shrestha*, St. Xavier's College in collaboration with ICTP Physics

11-13 Oct 2023: Workshop on Basics of *Quantum Espresso (QE)* Software, St. Xavier's College

October 20 2023: XI WWlet Wavelets and Applications in Signal and PDEs (online), SBMAC, Brazil | [link](#)

SUMMER SCHOOLS.....

Summer School on Gravitational-Wave Astronomy | [link](#)

1-12 July 2024

Participant-Online | [link](#)

ICTS, Bengaluru, India

Courses:

- The astrophysics of neutron stars and binaries
- Ultralight boson clouds around black holes
- Rapidly spinning neutron stars and emission mechanisms

TEACHING AND VOLUNTEER/LEADERSHIP EXPERIENCE

KU City College/School

Saraswatinagar, Chabahil, Kathmandu

Part-Time Teacher

2025-present

- Teaching Physics & Mathematics to NEB (Nepal Education Board) higher school students

Space Academy Institute

Narephant, Kathmandu

Tutor (Hybrid-mode)

2024-2025

- Taught BE (Civil Engineering) at Institute of Engineering (IOE), Nepal Physics I, Math I, Math II
 - Physics I: *Oscillation, Wave Motion, and Acoustics, Electricity and Magnetism, Quantum Theory and Quantum Mechanics*
 - Math I: *Differential Calculus, Integral Calculus, Vectors and Solid Geometry, Differential Equations*
 - Math II: *Linear Algebra, Vector Functions and Functions of Several Variables, Infinite Series, Complex Numbers and Functions*
- Taught high school Physics and Math

Physics Demonstration by SXPC-Nepal in Set Exhibition

Kathmandu, Nepal

Volunteer and Demonstrator

February 2, 2024

- Conducted a physics exhibition showcasing practical demonstrations of concepts such as gyroscopic precession, Chladni patterns, the Brachistochrone problem, and ferrofluid properties.

The Renaissance Library | [link](#)

Sindhuli, Nepal

President

2022-2024

- Established Mini library in three public places (Bhiman Hospital, Bhiman Police Area and Ward Municipality office) as 1 year plan but was extended later for another year.
- Conducted a cybersecurity awareness program in presence of field expertise as an event speaker, and secondary level students of all school in Bhiman area as a target audience.

SKILLS

Programming:: Python, R, Javascript, CPP, Latex(overleaf), Bash(Linux), SQL

Tools: Numpy, Scikit-Learn, Tensorflow, Lightkurve, Pandas, Astroquery, CAMB, DS9, TOPCAT, AstrolmageJ, Scipy

ADDITIONAL INFORMATION

Languages: English (**IELTS overall score: 7.0**), Nepali

Interests: Cosmology, Coding, Exploring, Travelling, Music, Teaching, Philosophy, Playing guitar in future

REFERENCES

Available upon request.